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Institute of Information Technology (IIT)

Project
On

“EduRide Transport Management System”

Submitted By

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for the partial fulfillment of the degree of Executive
Master in Information Technology*



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Acknowledgement

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Declaration

This is to declare that this project is my original work. No part of this work has been submitted elsewhere, partially or entirely, for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. The plagiarism policy, as stated by the supervisor, has been maintained.

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Signature Page

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Abstract

The EduRide Transport Management System is a comprehensive software solution designed to streamline campus transportation operations. By replacing manual processes with an automated, user-centric platform, the system aims to reduce administrative burdens, enhance resource optimization, and improve service quality for all stakeholders, including students, teachers, administrators, staff, and drivers. Built using the MERN stack (MongoDB, Express.js, React.js, Node.js), the application features role-specific dashboards that facilitate route management, real-time tracking, payments, and communication. This report outlines the project's motivation, description, technical strategy, and implementation details, highlighting its potential to deliver efficiency, reliability, and user empowerment in educational environments.

Keywords: EduRide Transport Management System, Campus Transportation, Transport Management System, Real-time Vehicle Tracking, MERN Stack, MongoDB, Express.js, React.js, Node.js.

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Chapter 1

Introduction

1.1 Background

Transportation management is a crucial operational component of educational institutions such as universities and colleges. Every day, students, teachers, staff, and drivers depend on campus transportation services to ensure smooth commuting between residential areas and academic premises. However, many institutions still rely on manual systems for managing transport services, which creates inefficiency, confusion, scheduling conflicts, lack of transparency, and increased administrative workload. With the advancement of modern web technologies, it has become possible to develop intelligent transport management platforms that provide real-time tracking, automated scheduling, digital enrollment, and seamless communication among all stakeholders. The EduRide Transport Management System has been developed to address these challenges by offering a smart, reliable, and scalable campus transport solution.

1.2 Motivation

The motivation behind EduRide arises from the difficulties faced in traditional campus transport systems. Manual operations require excessive paperwork, physical approvals, and constant communication between transport offices and users.

EduRide enhances the campus transport experience by:

- Reducing administrative hassle.
- Digitizing route and slot allocation.
- Providing real-time vehicle tracking.
- Ensuring transparency in enrollment and payments.

- Improving communication through complaints and feedback.
- Optimizing vehicle and driver resource utilization.

The project is designed to ensure efficiency, reliability, and improved service quality for students, teachers, administrators, staff members, and drivers.

1.3 Problem Statement

Manual transport management systems in educational institutions suffer from several limitations:

- Difficulty in managing multiple routes and time slots.
- Lack of live vehicle monitoring.
- Inefficient driver and vehicle assignment.
- Poor communication between transport authority and users.
- No automated complaint or feedback mechanism.
- Absence of digital payment notification and receipt generation.
- Increased chances of human error and delays.

Therefore, there is a strong need for an automated and centralized transport management system.

1.4 Project Objectives

The EduRide Transport Management System aims to improve the efficiency of campus transportation by introducing a centralized and automated management platform. The system focuses on reducing manual administrative work and providing a structured digital solution for managing transport operations. It provides role-based dashboards for different users such as administrators, drivers, and passengers to ensure organized access and management of transport services. The system also supports dynamic route and slot selection, allowing users to enroll in available transport routes conveniently.

Additionally, EduRide enables real-time vehicle tracking, helping users monitor vehicle movement and improving service transparency. It also assists administrators in managing driver duties effectively and ensures that users receive notifications regarding enrollment and payment updates. Through these features, the system enhances the overall efficiency and reliability of campus transportation services.

1.5 Scope of the Project

EduRide manages all aspects of campus transportation services. The system is intended for:

- Students who sign up for transport services.
- Teachers and staff in need of commuting assistance.
- Drivers who offer transport duty and live tracking.
- Administrators who oversee routes, vehicles, drivers, and users.

The system guarantees centralized oversight and access based on user roles.

1.6 Significance of the Project

The EduRide Transport Management System contributes to improving the overall efficiency and organization of campus transportation services. By introducing a digital platform for managing transport operations, the system helps reduce manual administrative tasks and simplifies the management of routes, vehicles, users. The system also enhances the reliability and efficiency of transport services by enabling structured route management, driver duty allocation, automated processes. In addition, EduRide improves user satisfaction by providing a convenient and user-friendly interface for transport enrollment, notifications, and service updates. Furthermore, the system supports better utilization of transport resources, ensuring that routes, and schedules are managed more effectively. Through these improvements, EduRide promotes the adoption of modern digital transport services within universities, efficient transportation environment.

Chapter 2

Literature Review

2.1 Overview

Transport management systems have been widely adopted in modern smart campuses. Many institutions use digital systems for tracking buses, managing routes, and providing transport-related services.

A transport management system typically includes:

- Vehicle scheduling.
- Route optimization.
- Driver assignment.
- GPS-based tracking.
- User enrollment.
- Payment processing.
- Complaint and feedback support.

EduRide follows these principles while focusing on university campus needs.

2.2 Research Gap

Despite the increasing adoption of digital technologies in transportation systems, many institutional transport management frameworks remain limited in scope and functionality. Existing approaches primarily focus on isolated operational tasks such as payment processing, route listing, or driver assignment, rather than offering a fully integrated transport management ecosystem. In traditional campus environments, transport administration is still dependent on manual procedures including paper-based payment

verification, manual route allocation, and informal complaint handling mechanisms [1], [2]. These processes introduce inefficiencies, increase administrative workload, and create a high probability of human error.

Although some modern systems have attempted to digitize transport management through web-based platforms, they often fail to address the comprehensive needs of a university transport environment. Most existing solutions lack real-time vehicle monitoring capabilities, which are critical for improving operational transparency, passenger safety, and service reliability [1], [2]. Additionally, many systems do not implement role-based access control structures, resulting in limited usability for different stakeholders such as administrators, drivers, and passengers who require distinct operational interfaces and permissions [3], [4].

Another significant limitation observed in current literature and practical implementations is the absence of structured route-slot enrollment mechanisms tailored for campus transportation. Many transport platforms operate with static route assignments rather than dynamic slot-based enrollment systems that allow users to register for specific routes and schedules efficiently. Furthermore, existing systems rarely incorporate integrated communication modules for delivering transport notices, operational updates, or traffic law awareness to users within the same platform.

From a research perspective, the major gap lies in the lack of a comprehensive, university-focused transport management framework that integrates administrative control, passenger services, operational monitoring, and communication functionalities into a unified digital system. Current solutions are fragmented, focusing on single functionalities rather than providing a holistic management environment capable of supporting large academic institutions.

2.4 Conclusion

The review emphasizes the need for an integrated campus-specific transport management system to replace inefficient manual processes. EduRide fulfills this need by offering a centralized and scalable solution built with the MERN Stack. It streamlines route management, bus allocation, user coordination, and complaint handling through a secure digital platform. Overall, the system enhances operational efficiency and improves.

Chapter 3

System Analysis and Requirements

3.1 Requirement Analysis

EduRide is designed as a role-based web system with five primary user types:

- Admin.
- Student.
- Teacher.
- Staff.
- Driver.

Each role has specific permissions and functionalities.

3.2 Functional Requirements

Admin, Student, Teacher, Staff, Driver are five actors in this system.

Admin can:

- Assign routes and slots.
- Approve/ Pending/ Reject slot requests.
- Manage vehicles and users.
- Track vehicles in real time.
- Upload traffic law information.
- Manage Users.
- Publish notices.
- Handle complaints and give feedback.
- Monitor and verify student payments.

Students can:

- Select routes and slots.
- Make payments.
- Download payment receipts.
- Track vehicles live.
- Submit complaints.
- Enable location.
- Get complain feedback.
- View notices.
- Download student transport card.

Teachers can:

- Select routes and slots.
- Track vehicles.
- Enable location.
- Submit complaints.
- Receive complain feedback.
- View notices.

Staff can:

- Track vehicles.
- View notices.
- Select routes and slots.
- Enable location.
- Receive complain feedback.
- Submit complaints.

Drivers can:

- View assigned duty schedule.
- Enable live location sharing.
- View notices.
- Verify Student Payment.
- View traffic laws.

3.3 Non-Functional Requirements

The EduRide Transport Management System must satisfy several important system requirements to ensure effective and reliable operation. The system should be scalable, allowing it to handle an increasing number of users, vehicles, and transport routes as the university grows. It must also maintain a high level of reliability, ensuring that the system remains stable and operational without frequent failures. Security is another critical requirement, as the system needs to protect user data, payment information, and administrative controls from unauthorized access. Furthermore, EduRide should support real-time responsiveness so that users can receive immediate updates regarding vehicle tracking, route information, and transport notifications. The system must also provide a user-friendly interface, enabling students, staff, and administrators to interact with the platform easily and efficiently. Finally, the system should ensure database consistency, guaranteeing that all transport-related data such as routes, payments, and user information remain accurate, synchronized, and properly maintained within the database.

3.4 System Feasibility

- **Technical Feasibility:** MERN Stack provides full support for real-time scalable web applications. It enables seamless integration between the frontend and backend using JavaScript across the entire development process. With MongoDB's flexible database

structure, the system can efficiently manage large volumes of dynamic transport data. Node.js and Express.js ensure fast server-side processing and API handling, while React.js delivers an interactive and responsive user interface. This combination makes the application highly scalable, maintainable, and suitable for future expansion.

- **Economic Feasibility:** The system is cost-effective as it uses open-source tools. It eliminates the need for expensive proprietary software licenses and reduces overall development costs. The availability of a large developer community ensures continuous updates, security improvements, and technical support. Additionally, open-source technologies allow easy customization and scalability without significant financial investment.
- **Operational Feasibility:** The platform is easy to use for all campus stakeholders. It provides a simple and intuitive interface that requires minimal technical knowledge. Role-based dashboards ensure that students, staff, drivers, and administrators can quickly access relevant features. Clear navigation and responsive design make the system accessible from both desktop and mobile devices.

Chapter 4

System Design and UML Diagrams

4.1 Use Case Diagram

Use case diagrams give the non-technical view of the overall system. The following use cases were made from the scenario.

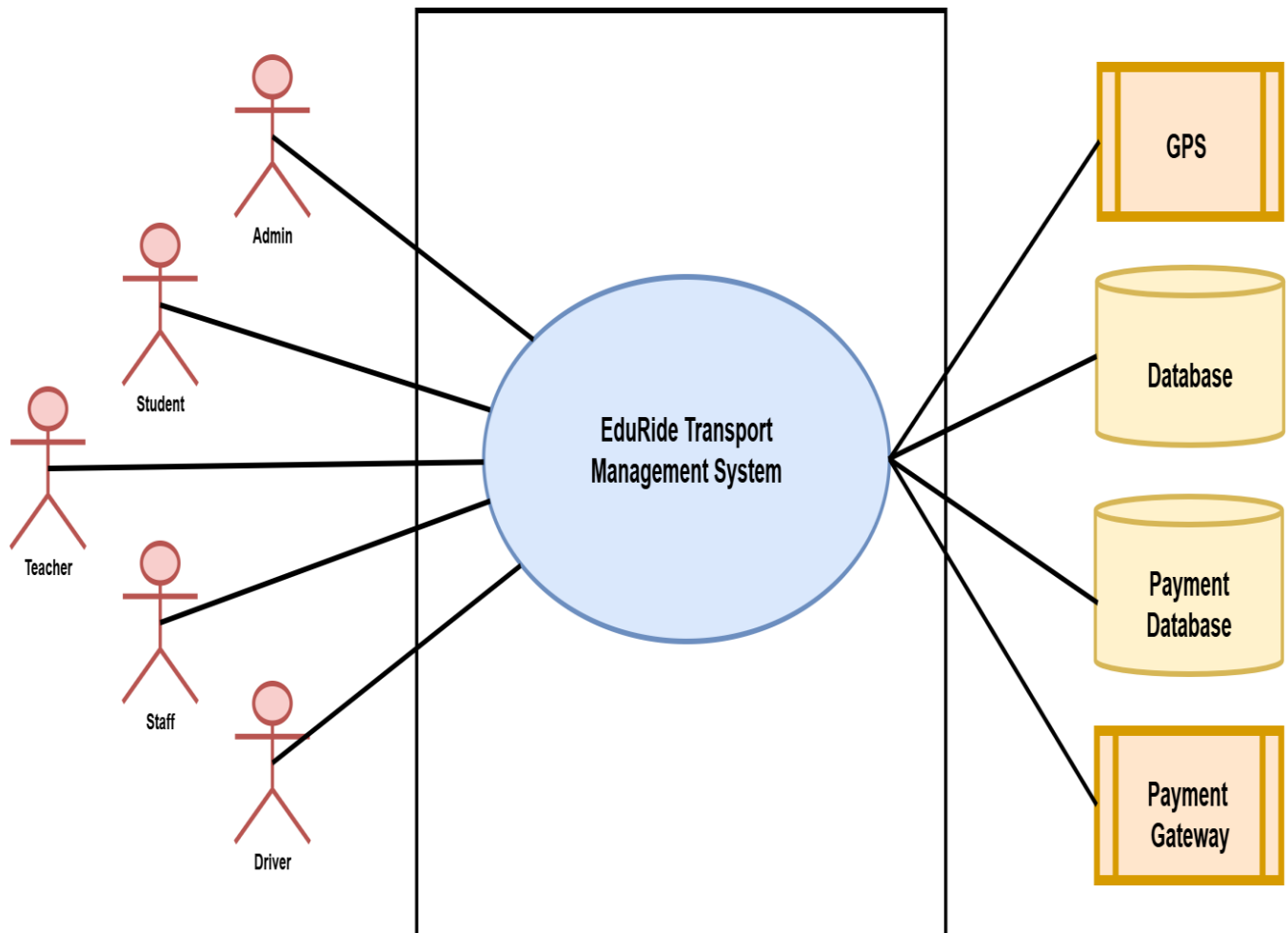


Figure 1: Level 0 Use Case Diagram.

4..1.1 Level 0 Use Case Diagram

It (Figure: 1) presents the overall use case and system architecture of the EduRide Transport Management System. It illustrates five main actors — Admin, Student, Teacher, Staff, and Driver — who interact directly with the centralized EduRide system according to their specific roles. The Admin is responsible for managing routes, vehicles, drivers, users, and overall system operations. Students, teachers, and staff use the platform for transport enrollment, route and slot selection, payment processing, and submitting complaints or feedback. Drivers access the system to view duty schedules and provide transport updates, including real-time location sharing.

At the core, the EduRide system processes all user requests and coordinates transport operations efficiently. The system integrates with a GPS module to collect real-time vehicle location data, enabling live tracking functionality. All operational information is stored in the main central database, while financial transactions are securely maintained in a separate payment database. Online payments are handled through an integrated payment gateway to ensure secure and reliable transactions.

Overall, the diagram represents a fully centralized and integrated transport management platform that combines user interaction, real-time GPS tracking, database management, and digital payment services into a unified and scalable system.

EduRide Transport Management System

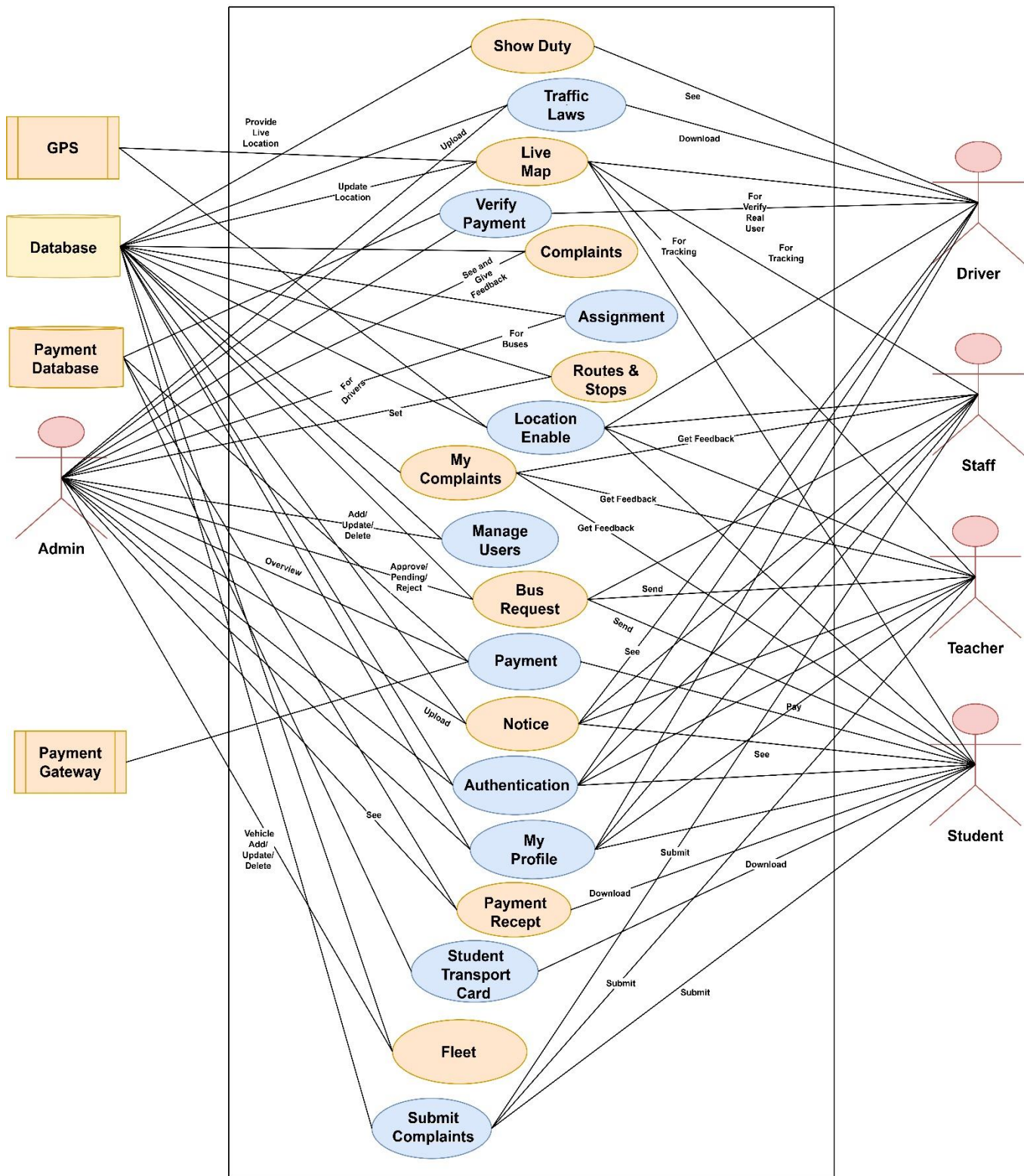


Figure 2: Level 1 Use Case Diagram.

4..1.2 Level 1 Use Case Diagram

The “EduRide Transport Management System” Level 1 Use Case Diagram (Figure:2) illustrates the functional structure and user interactions of the system. It defines the system boundary and shows how five main actors—Admin, Driver, Student, Staff, and Teacher—interact with various transport management features.

The Admin has full control over system operations, including user management, fleet management, route and driver assignments, payment verification, complaint handling, and notice publication. Students, Staff, and Teachers use the system to request transport services, make online payments, track buses in real time, view notices, and submit complaints. Students can also download transport cards and payment receipts after successful transactions. Drivers access duty schedules, assigned routes, traffic guidelines, and live map navigation.

The system integrates with external components such as GPS for real-time tracking, a Database for storing operational data, and a Payment Gateway with Payment Database for secure transaction processing and verification.

Overall, the diagram demonstrates structured role-based access control, real-time monitoring, secure payment processing, and centralized data management, ensuring efficient and transparent campus transportation services.

4.2 Data Flow Diagram

4.2.1 DFD Diagram Level 0

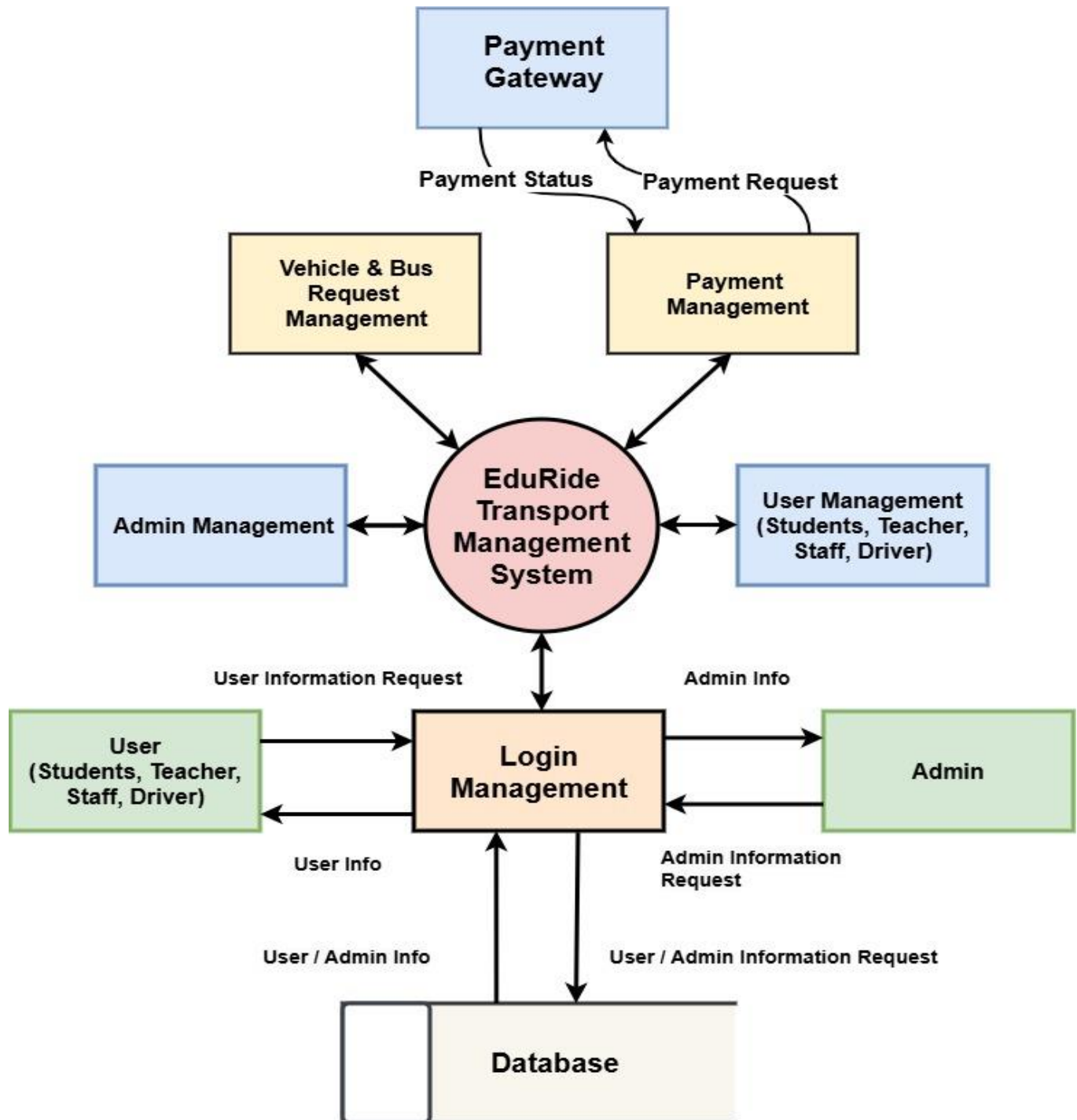


Figure 3: DFD Diagram Level 0.

The EduRide Transport Management System is a centralized, role-based platform developed to efficiently manage school or college transportation services. The system begins with a secure login management process where students, teachers, staff, drivers, and administrators authenticate themselves using valid credentials. These credentials are verified against a central database to ensure authorized access and maintain system security. After successful authentication, users are directed to their respective dashboards based on their assigned roles and permissions.

The core system functions as a central coordinator that integrates multiple management modules, including user management, admin management, vehicle and bus request management, and payment management. Users can request vehicles or bus services through the system, and administrators can manage and assign transport resources accordingly. The payment management module handles transport fees and securely communicates with an external payment gateway to process transactions. The gateway returns real-time payment status updates to ensure transparency and accuracy. Overall, the system ensures secure, efficient, and accountable transport operations through integrated authentication, centralized database control, vehicle allocation, and online payment processing.

4.2.2 DFD Level 1 (Admin)

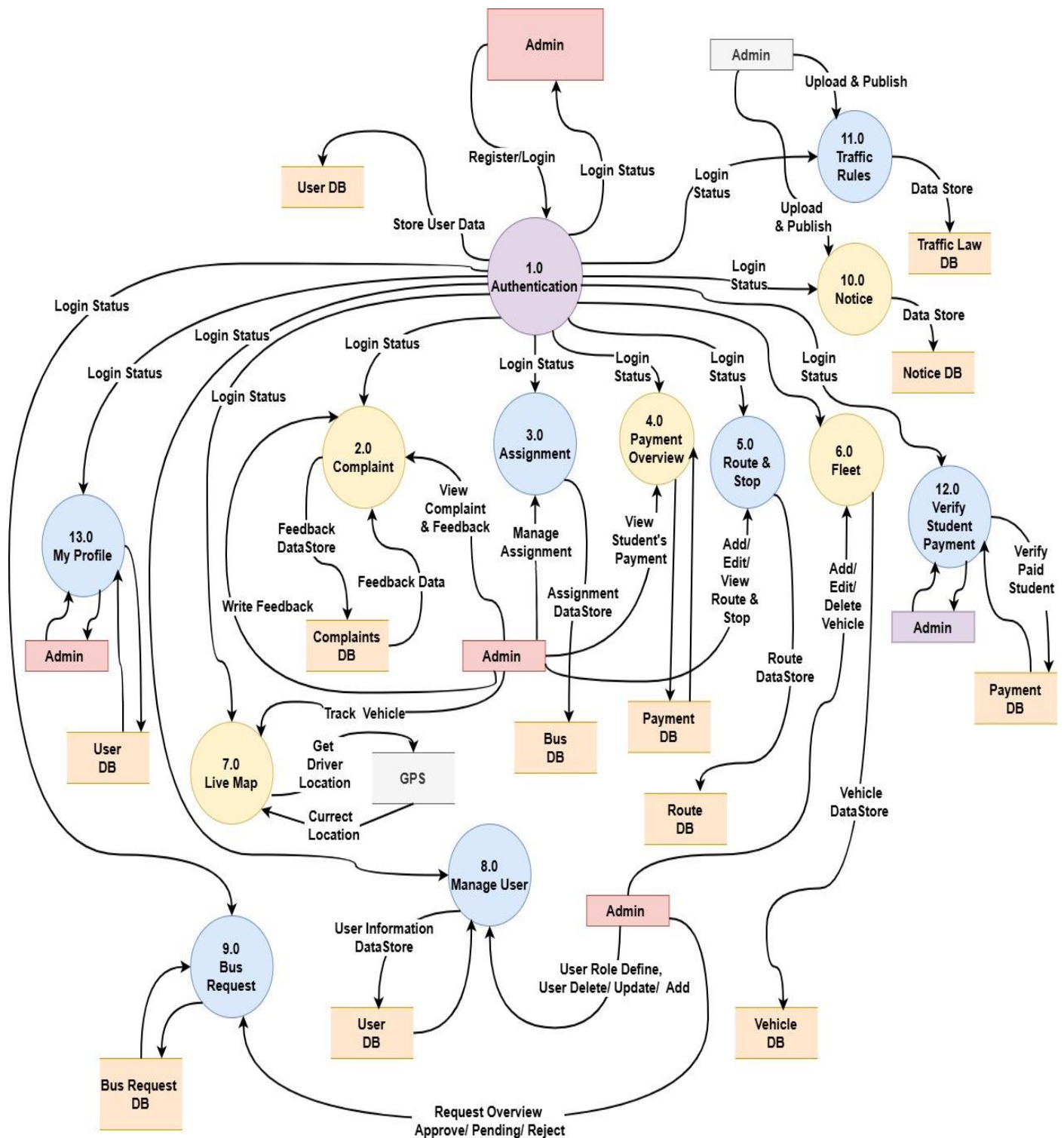


Figure 4: DFD Level 1 (Admin).

The diagram (Figure 4) represents the Admin side of the transport management system. First, the Admin logs in through the 1.0 Authentication module. The login information is verified using the User DB, and after successful authentication, the admin can access all management modules. The admin can manage complaints in the 2.0 Complaint module by viewing student complaints and giving feedback. All complaint data is stored in the Complaints DB. Through the 3.0 Assignment module, the admin manages bus assignments, which are stored in the Bus DB. In the 4.0 Payment Overview and 12.0 Verify Student Payment modules, the admin views payment records and verifies paid students using the Payment DB. The 5.0 Route & Stop module allows the admin to add, edit, and manage routes and stops, with data stored in the Route DB. The 6.0 Fleet module is used to add, edit, or delete vehicles, and the data is stored in the Vehicle DB. The admin can publish announcements through the 10.0 Notice module (stored in the Notice DB) and upload traffic rules through the 11.0 Traffic Rules module (stored in the Traffic Law DB). Using the 8.0 Manage User module, the admin can add, update, delete, or redefine user roles. The 9.0 Bus Request module allows the admin to review, approve, or reject bus requests from students. The 7.0 Live Map module enables the admin to track vehicles in real time using GPS.

Overall, the admin system controls and manages users, vehicles, routes, payments, complaints, notices, and real-time tracking to ensure smooth transport service operation.

4.2.3 DFD Level 1 (Teacher)

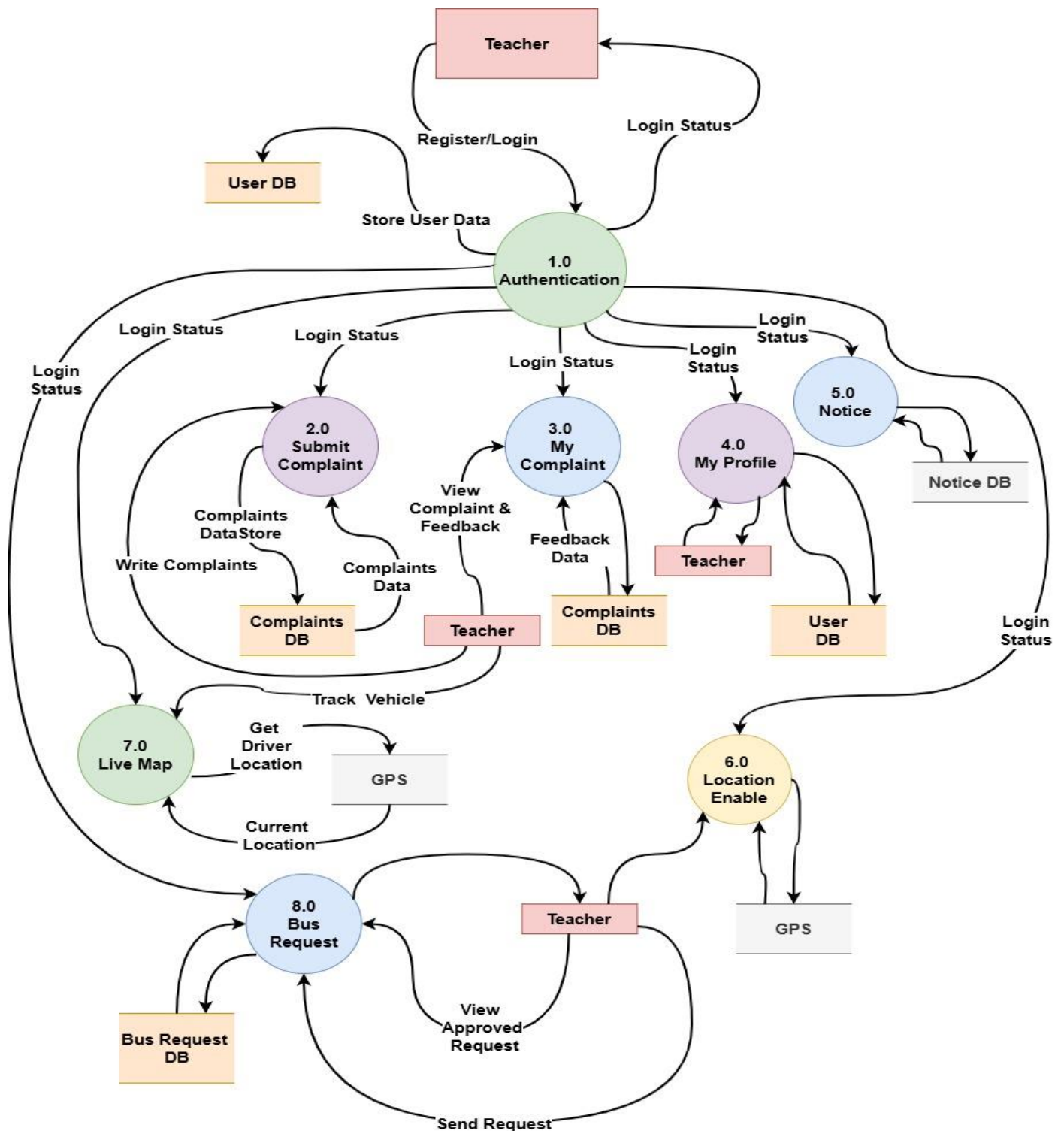


Figure 5: DFD Level 1 (Teacher).

The Level-1 DFD (Teacher) of the “EduRide Transport Management System” (Teacher Module) illustrates the interaction between a teacher member and the system, as well as the data flow among processes and databases. The process of working starts with Authentication, during which teacher register or log in and their credentials are confirmed via the User Database. Once logged in successfully, teacher members can manage their profiles, view notices from the Notice Database, and send bus requests that are saved in the Bus Request Database.

In case of any issues, teacher can file complaints which will be recorded in the Complaints Database. They can subsequently check for feedback or updates on the status. Through the Live Map module, which gathers location data from GPS once location permission is activated, the system also offers real-time vehicle tracking.

4.2.4 DFD Level 1 (Student)

The (Figure 6: DFD Diagram Level 1 (Student)) allows a **Student** to register or log in through the Authentication module, where user data is stored in the User DB. After successful login, the student can access different services. Students can manage their profile, submit and view complaints, make transport payments, and receive payment slips and transport card updates. They can also view notices from the system. By enabling location services, students can track the bus in real time using GPS and the live map. Additionally, they can send bus requests and check their approval status.

Overall, the system combines authentication, complaint handling, payment processing, notice updates, GPS tracking, and bus request management to support student transport services efficiently.

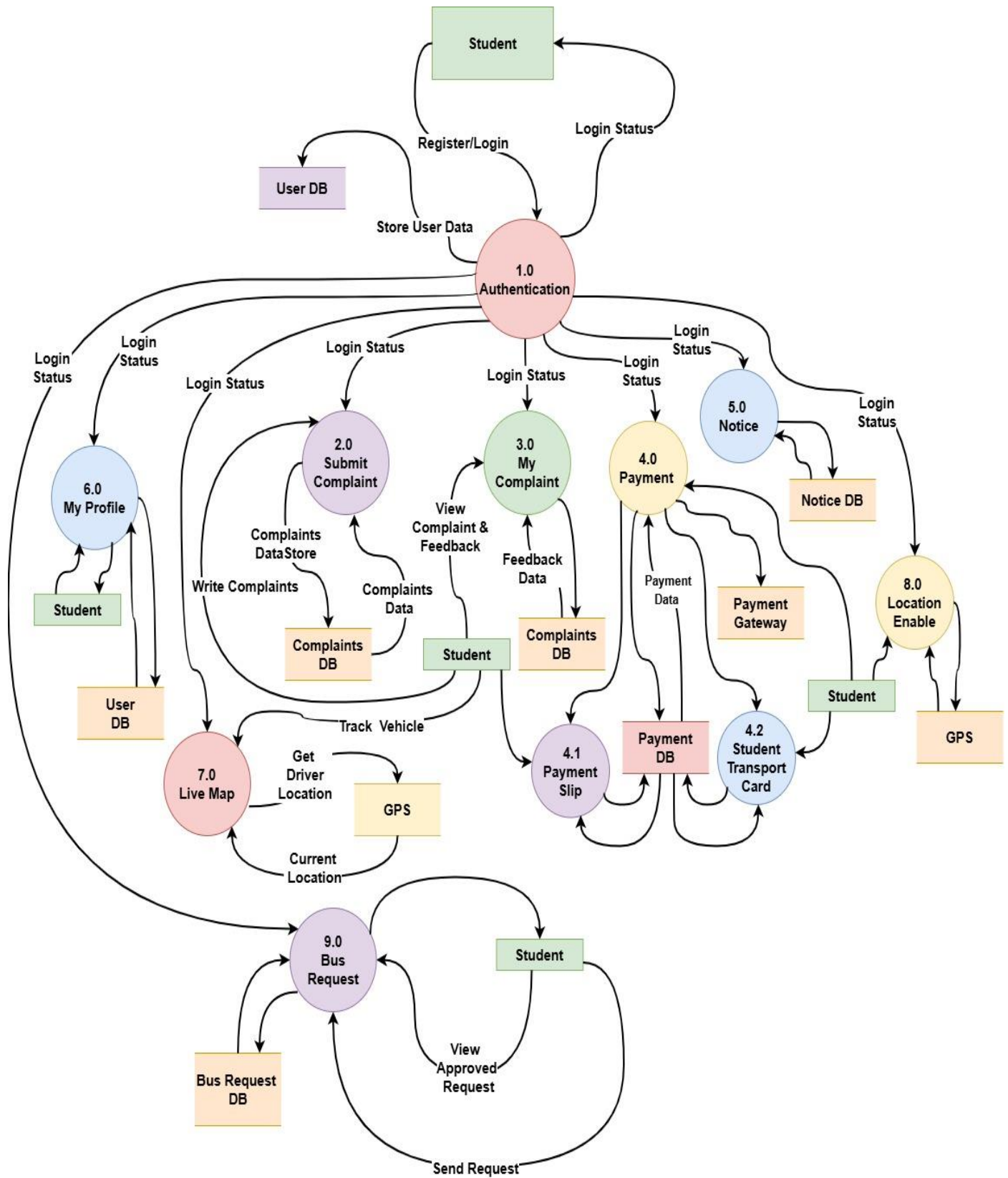


Figure 6: DFD Level 1 (Student).

4.2.5 DFD Level 1 (Driver)

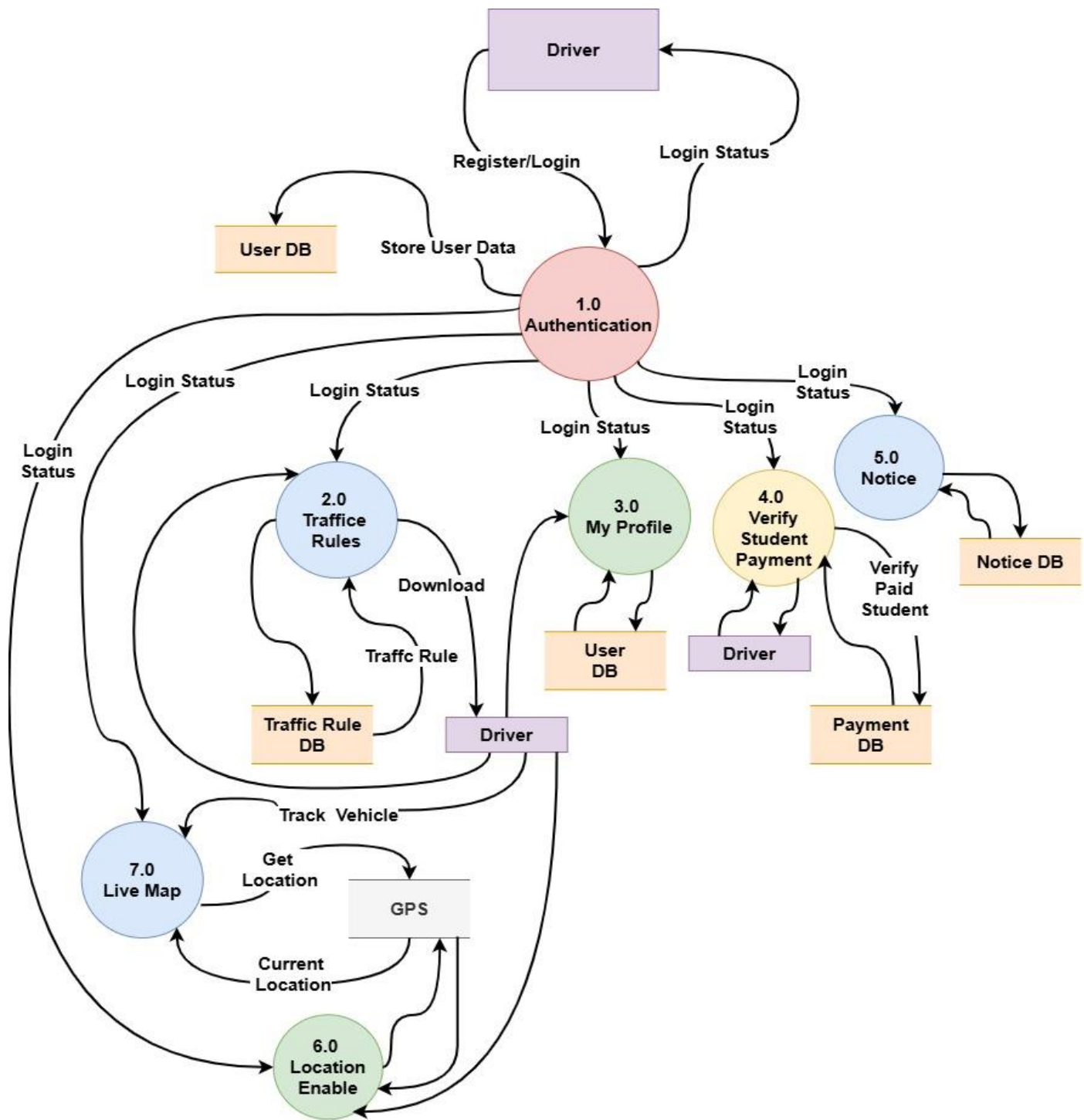


Figure 7: DFD Level 1 (Driver).

The (Figure 7: DFD Diagram Level 1 (Driver)) allows the Driver to log in through the Authentication module, where user information is stored in the User DB. After successful login, the driver can access different services. The driver can view traffic rules, manage profile information, verify student payments, and check notices from their respective databases. When location is enabled, the system uses GPS to provide real-time tracking through the live map.

Overall, the system combines authentication, database management, payment verification, and GPS tracking to support driver activities efficiently.

4.2.6 DFD Diagram Level 1 (Staff)

The (Figure:8) Level-1 DFD (Staff) of the “EduRide Transport Management System” (Staff Module) describes how a staff member interacts with the system and how data flows between processes and databases. The working process begins with Authentication, where staff register or log in and their credentials are verified through the User Database. After successful login, staff can manage their profile, view notices from the Notice Database, and send bus requests which are stored in the Bus Request Database.

If any issue arises, staff can submit complaints that are stored in the Complaints Database and later view feedback or status updates. The system also provides real-time vehicle tracking through the Live Map module, which collects location data from GPS after location permission is enabled.

Overall, the diagram shows structured data processing, secure user authentication, database interaction, and GPS integration to ensure efficient and transparent transport service management.

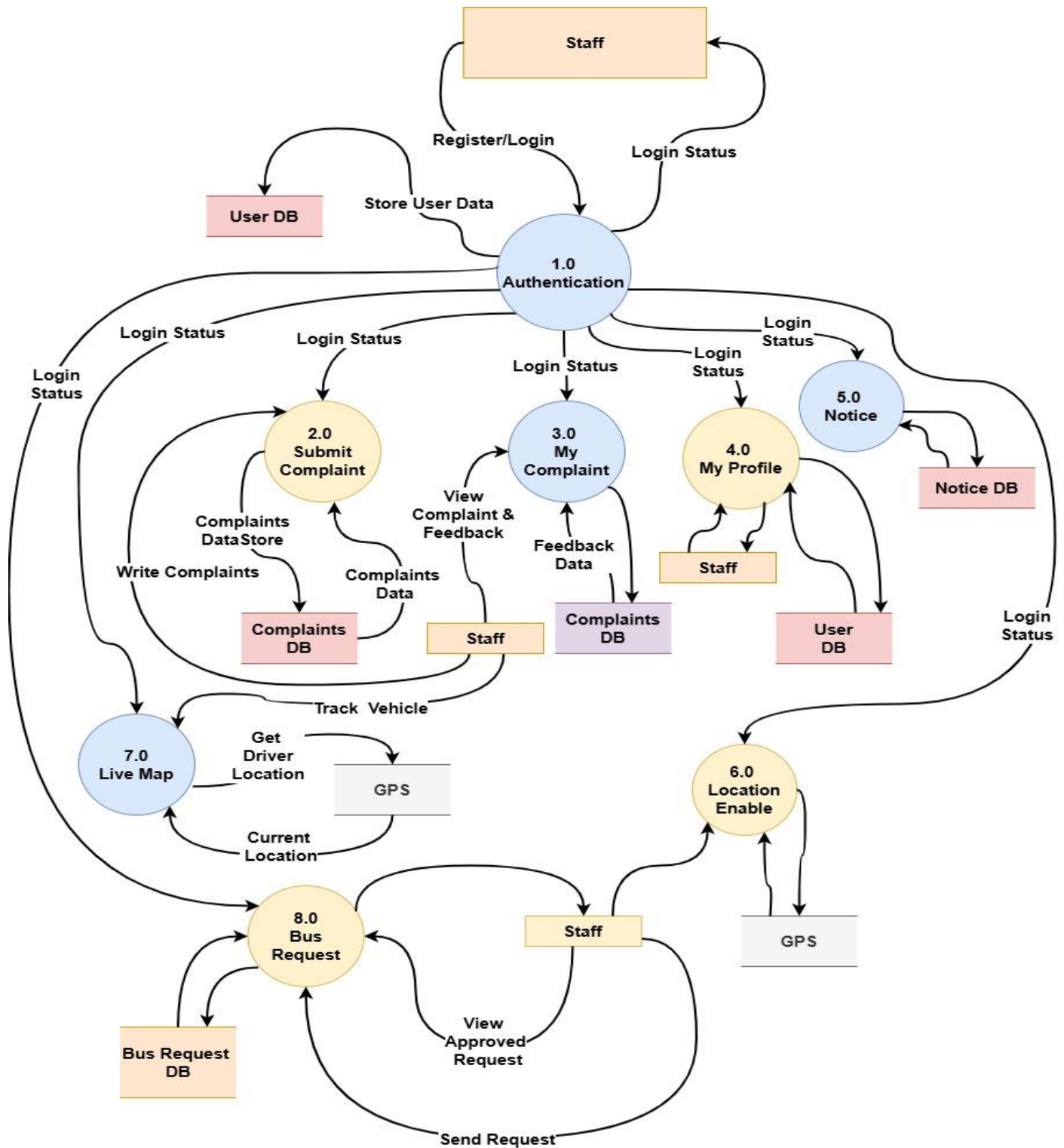


Figure 8: DFD Level 1 (Staff).

Chapter 5

Implementation

5.1 Tools and Technology

Programming Languages:

- Backend Development: Node.js, Express.js (MERN Stack)
- Frontend Development: React.js, HTML, CSS, Tailwind CSS, JavaScript
- Database: MongoDB

Integrated Development Environment (IDE):

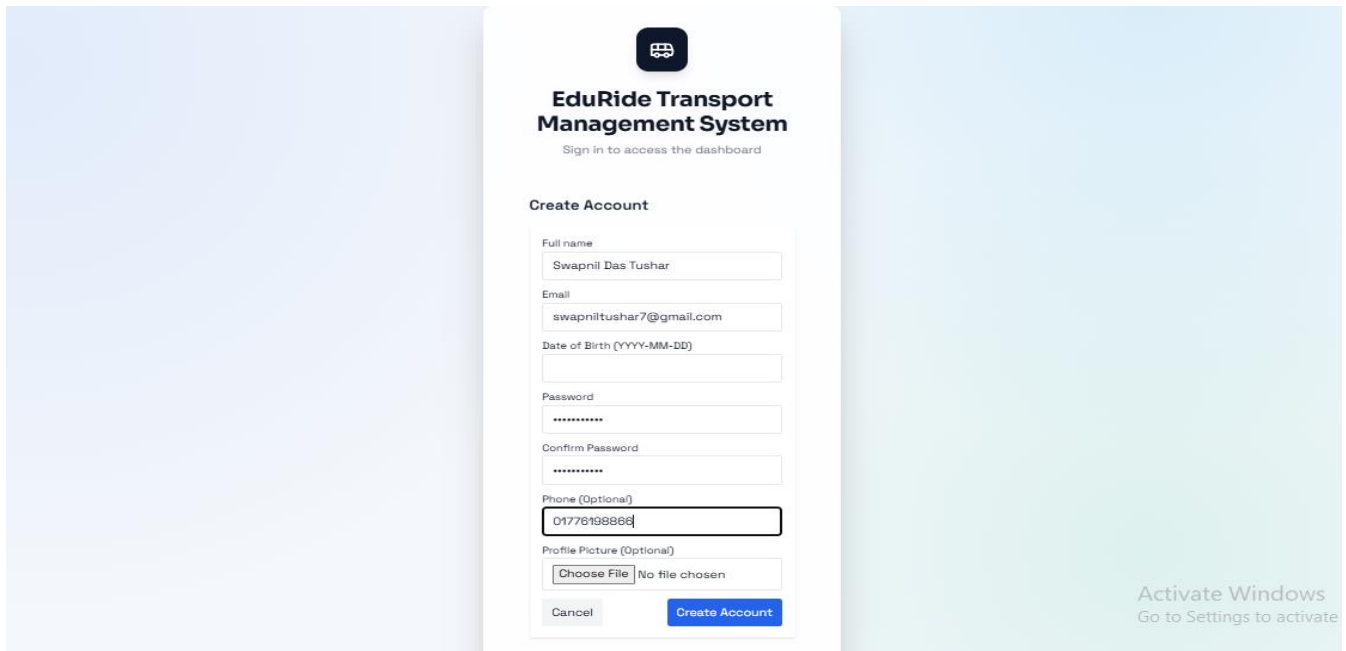
- Visual Studio Code

Version Control:

- Git, GitHub

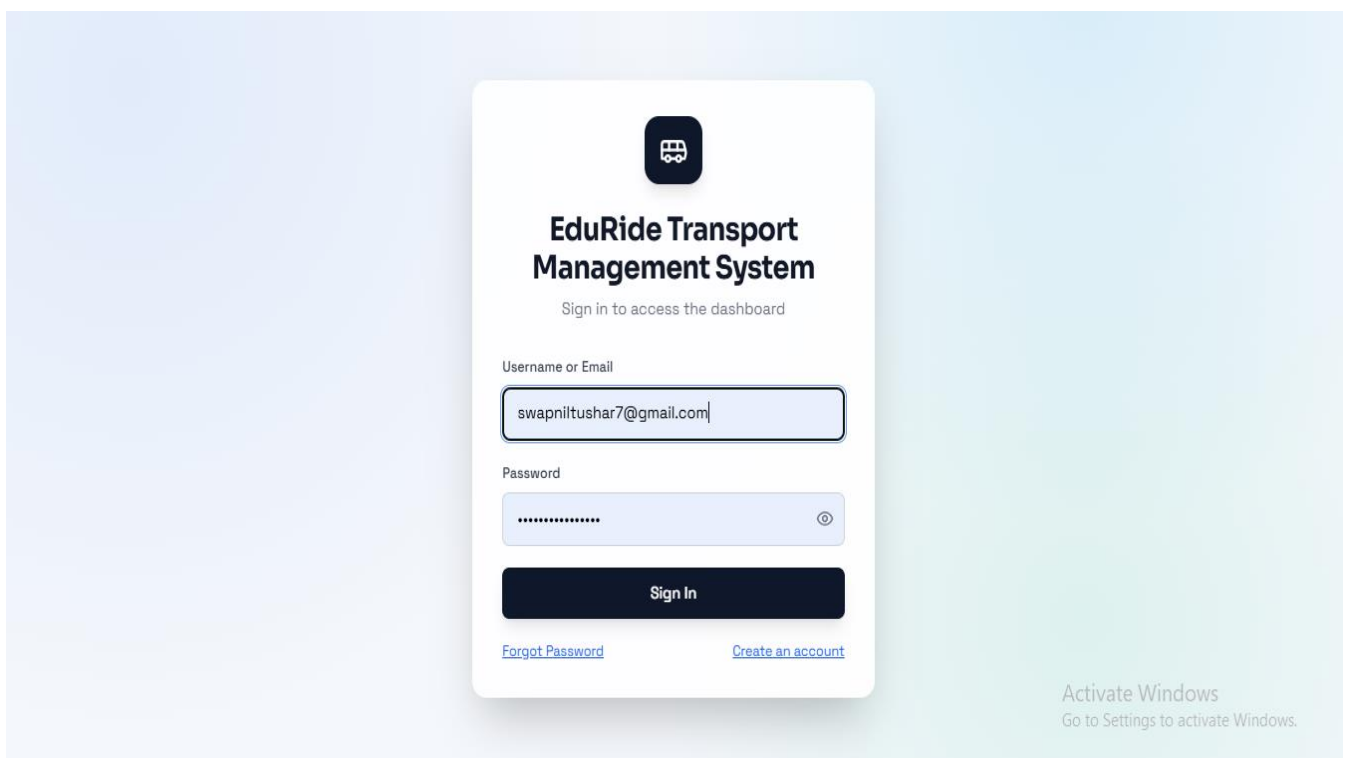
5.2 Project View

Signup and Login Interface



The image shows a web interface for the 'EduRide Transport Management System'. At the top, there is a dark blue icon of a bus. Below it, the title 'EduRide Transport Management System' is displayed in bold, followed by the subtitle 'Sign in to access the dashboard'. The main section is titled 'Create Account' and contains several input fields: 'Full name' (with the text 'Swapnil Das Tushar'), 'Email' (with the text 'swapniltushar7@gmail.com'), 'Date of Birth (YYYY-MM-DD)' (empty), 'Password' (masked with dots), 'Confirm Password' (masked with dots), 'Phone (Optional)' (with the text '01778198868'), and 'Profile Picture (Optional)' (with a 'Choose File' button and the text 'No file chosen'). At the bottom of the form are two buttons: 'Cancel' and 'Create Account'. On the right side of the interface, there is a watermark that says 'Activate Windows Go to Settings to activate'.

Figure 9: Signup Interface (For All Users).



The image shows a web interface for the 'EduRide Transport Management System'. At the top, there is a dark blue icon of a bus. Below it, the title 'EduRide Transport Management System' is displayed in bold, followed by the subtitle 'Sign in to access the dashboard'. The main section contains two input fields: 'Username or Email' (with the text 'swapniltushar7@gmail.com') and 'Password' (masked with dots). Below these fields is a dark blue button labeled 'Sign In'. At the bottom of the form are two links: 'Forgot Password' and 'Create an account'. On the right side of the interface, there is a watermark that says 'Activate Windows Go to Settings to activate Windows'.

Figure 10: Login Interface (For All Users).

Admin Dashboard Interface

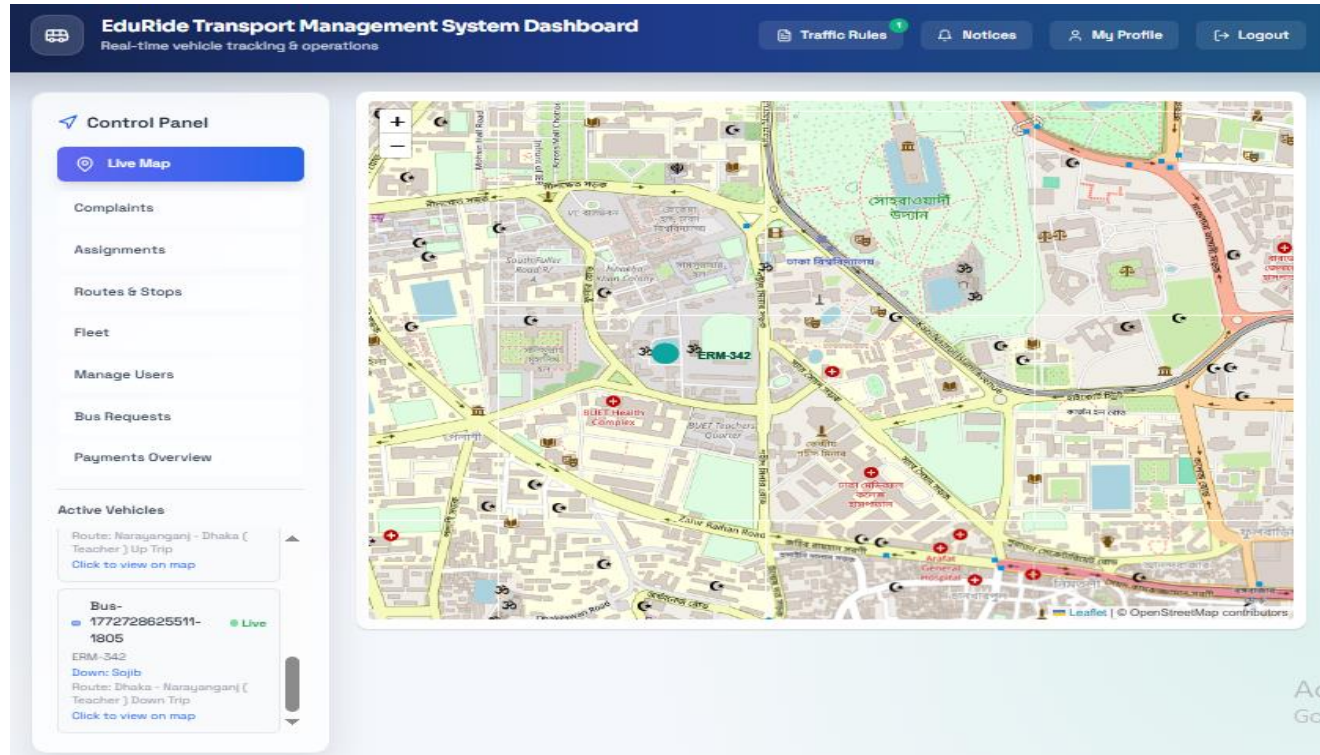


Figure 11: Admin Dashboard (Live Map for Tracking).

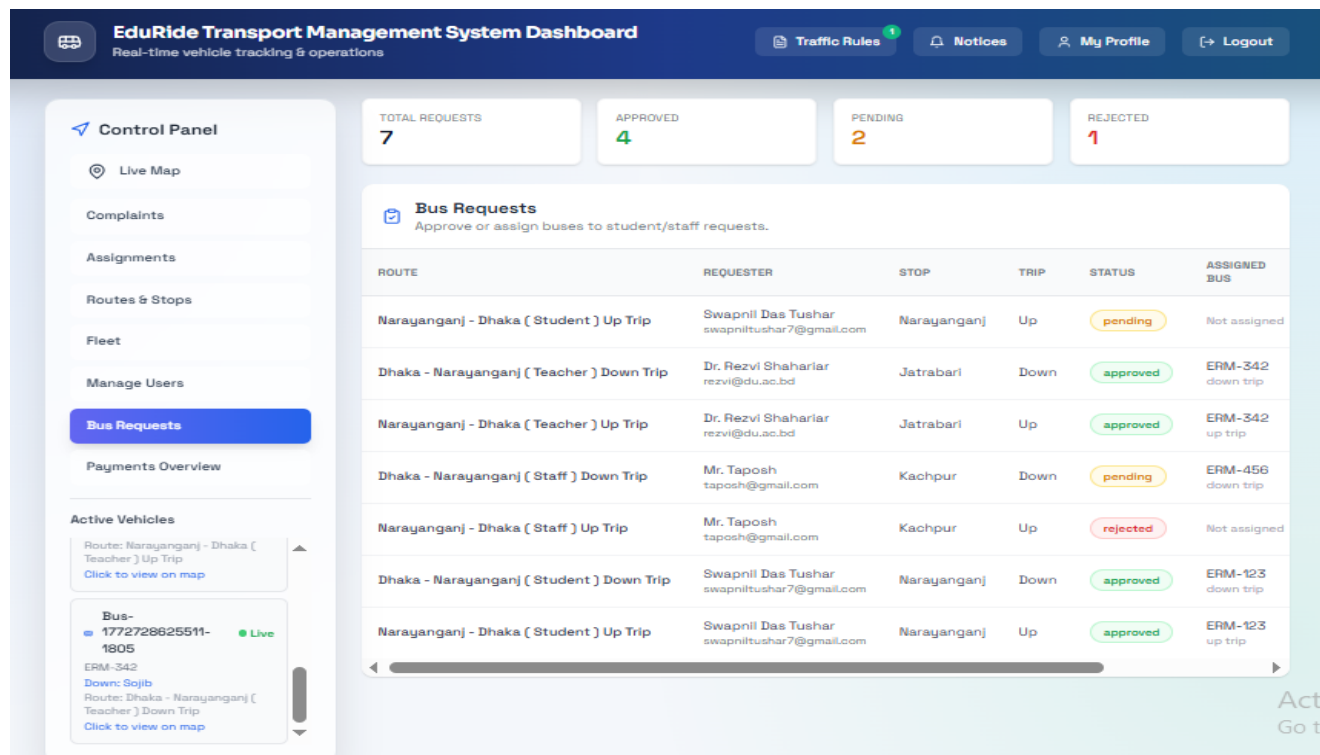


Figure 12: Admin Dashboard (Bus Request).

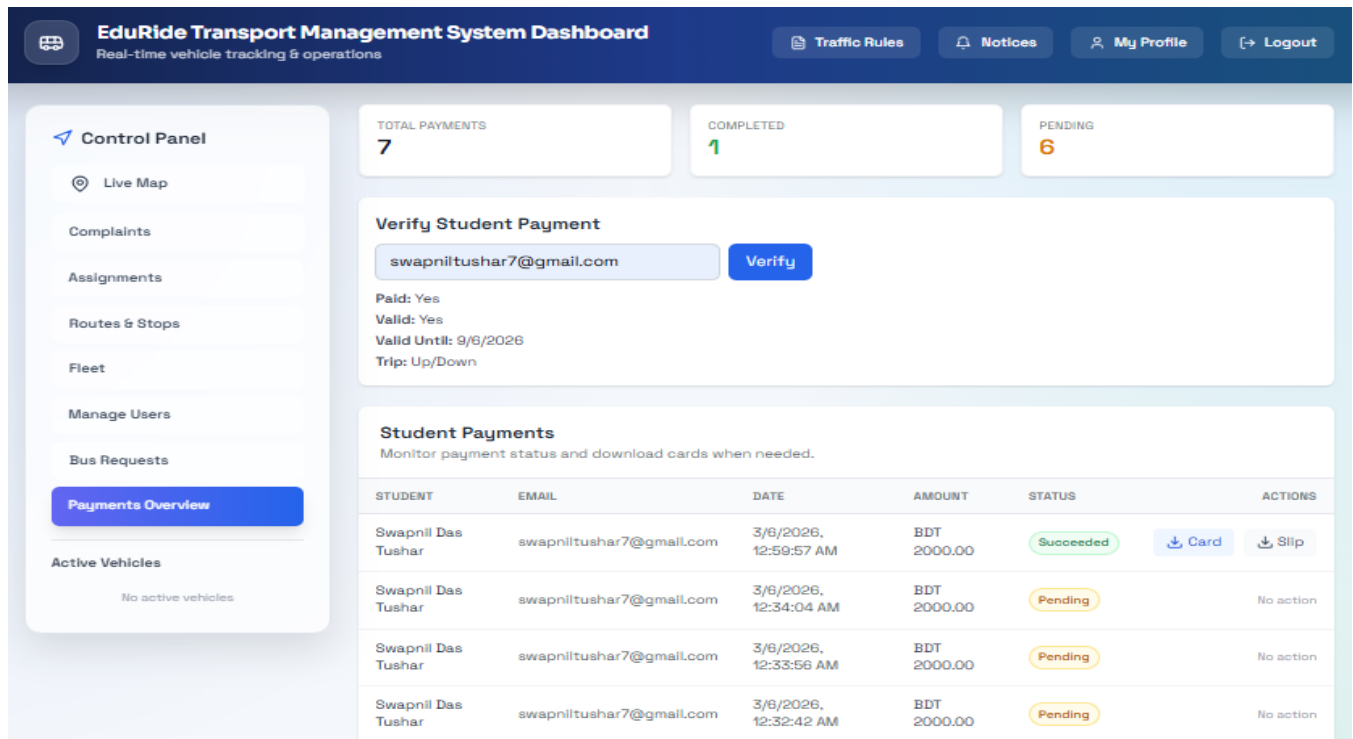


Figure 13: Admin Dashboard (Payment Overview).

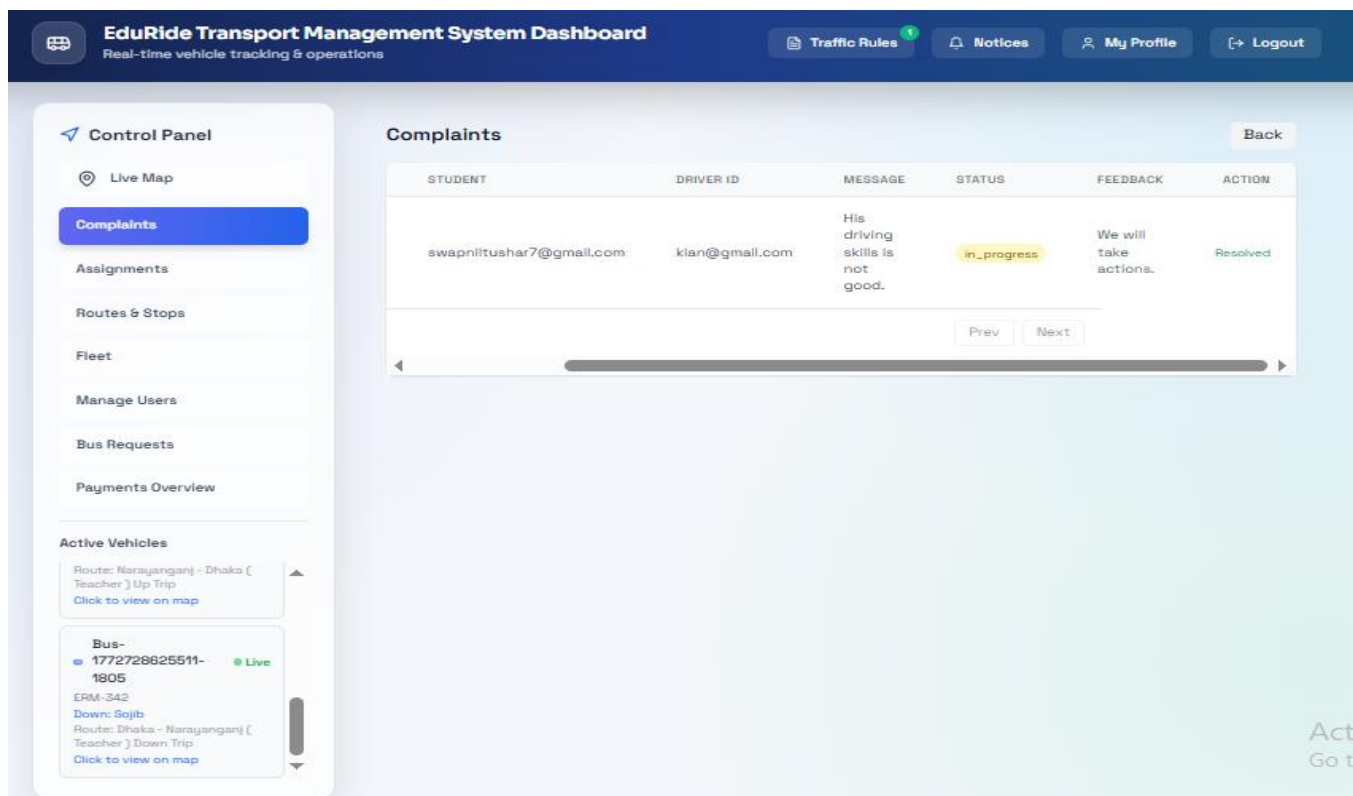


Figure 14: Admin Dashboard (Complaints).

Student Dashboard Interface

The screenshot shows the 'Submit Complaint' page of the EduRide Transport Management System. The header includes the system name, a subtitle, and navigation links for Notices, My Profile, and Logout. A left sidebar contains a 'Control Panel' with links to Live Map, Enable My Location, Bus Requests, Submit Complaint, Payments & Card, My Complaints, and Active Vehicles. The main content area is titled 'Submit Complaint' and features a 'Back' button. It contains a form with a dropdown for 'Against Driver' (selected: Kian (kian@gmail.com)), a text area for 'Complaint Message' (containing 'His driving skills is not good.'), and 'Cancel' and 'Submit Complaint' buttons. An 'Activate Windows' watermark is visible in the bottom right corner.

EduRide Transport Management System Dashboard
Real-time vehicle tracking & operations

Notices My Profile Logout

Control Panel

- Live Map
- Enable My Location
- Bus Requests
- Submit Complaint
- Payments & Card
- My Complaints
- Active Vehicles
No active vehicles

Submit Complaint Back

Against Driver
Kian (kian@gmail.com)

Complaint Message
His driving skills is not good.

Cancel Submit Complaint

Activate Windows
Go to Settings to activate

Figure 15: Student Dashboard (Submit Complaint).

The screenshot shows the 'My Complaints' page of the EduRide Transport Management System. The header and sidebar are identical to Figure 15. The main content area is titled 'My Complaints' and features a 'Back' button. It displays a table with one complaint entry. The table has columns for ID, DRIVER, MESSAGE, STATUS, and FEEDBACK. The entry shows ID C-1772735399322-184, DRIVER kian@gmail.com, MESSAGE His driving skills is not good., STATUS open, and FEEDBACK No feedback yet. Below the table is a pagination bar showing 'Page 1 of 1' and 'Prev' and 'Next' buttons. An 'Activate Windows' watermark is visible in the bottom right corner.

EduRide Transport Management System Dashboard
Real-time vehicle tracking & operations

Notices My Profile Logout

Control Panel

- Live Map
- Enable My Location
- Bus Requests
- Submit Complaint
- Payments & Card
- My Complaints
- Active Vehicles
No active vehicles

My Complaints Back

ID	DRIVER	MESSAGE	STATUS	FEEDBACK
C-1772735399322-184	kian@gmail.com	His driving skills is not good.	open	No feedback yet

Page 1 of 1 Prev Next

Activate Windows
Go to Settings to activate

Figure 16: Student Dashboard (My Complaints).

EduRide Transport Management System Dashboard
 Real-time vehicle tracking & operations

[Notices](#)
[My Profile](#)
[Logout](#)

Control Panel

- Live Map
- Enable My Location
- Bus Requests**
- Submit Complaint
- Payments & Card
- My Complaints
- Active Vehicles

No active vehicles

Request a Scheduled Bus
 Choose a route and stop. Admins will review and assign you to the best available bus.

Route

Narayanganj - Dhaka (Student) Up Trip

Preferred Stop

Narayanganj (7:00 AM)

Trip

Up Trip

Additional Notes

Near NRB Bank.

Request submitted successfully.

Submit Request

Figure 17: Student Dashboard (Bus Request).

EduRide Transport Management System Dashboard
 Real-time vehicle tracking & operations

[Notices](#)
[My Profile](#)
[Logout](#)

Control Panel

- Live Map
- Enable My Location
- Bus Requests
- Submit Complaint
- Payments & Card**
- My Complaints
- Active Vehicles

No active vehicles

Pay Semester Transport Fee
 Amount: BDT 2000.00 (covers 6 months)

Semester

Semester

Pay Semester

Payment History
 Completed payments stay valid for 6 months from payment date.

STUDENT
69a97bd330010436536cfc5c

SUCCEEDED
1

PENDING
6

Date	Amount	Trip	Valid Until	Status	Actions
3/6/2026, 12:59:43 AM	BDT 2000.00	Up/Down	9/6/2026	Succeeded	Card Slip
3/6/2026, 12:34:04 AM	BDT 2000.00	Up/Down	9/6/2026	Pending	No action
3/6/2026, 12:33:56 AM	BDT 2000.00	Up/Down	9/6/2026	Pending	No action
3/6/2026, 12:32:42 AM	BDT 2000.00	Up/Down	9/6/2026	Pending	No action
3/6/2026, 12:32:10 AM	BDT 2000.00	Up/Down	9/6/2026	Pending	No action

Figure 18: Student Dashboard (Payment & Card).

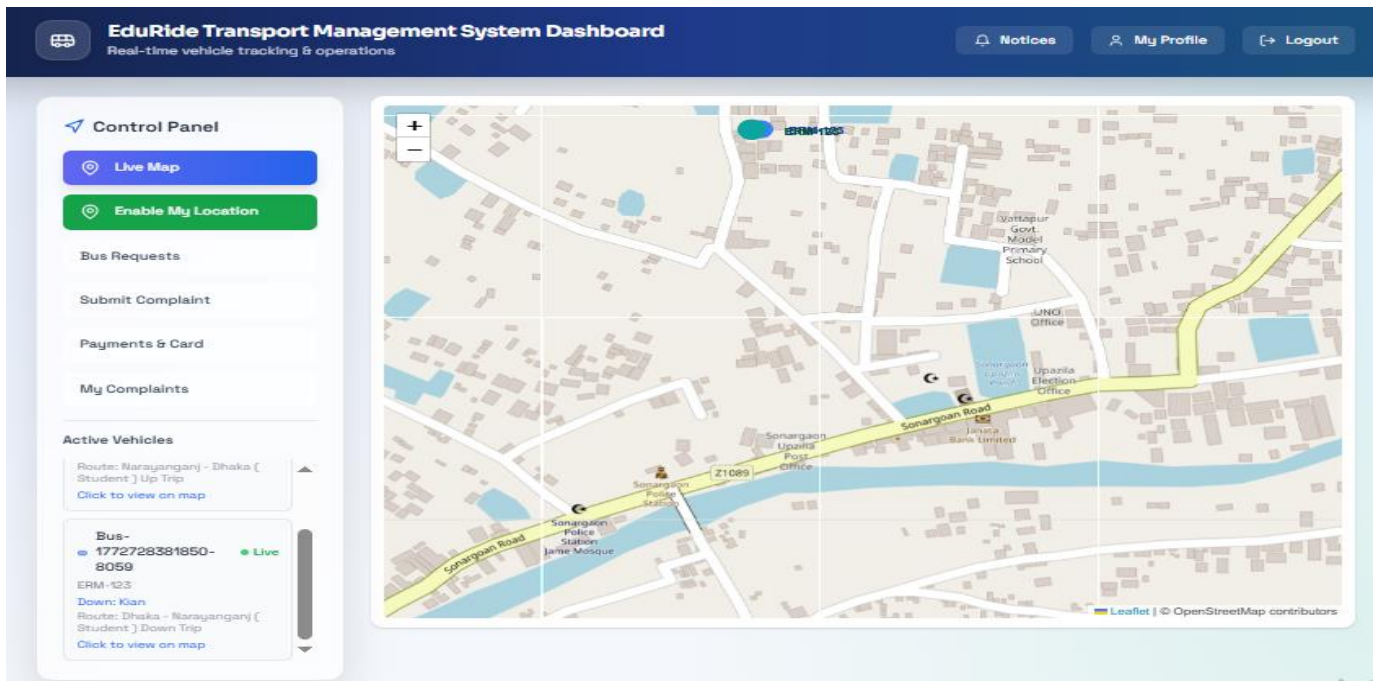


Figure 19: Student Dashboard (Active Vehicle).

Teacher Dashboard Interface

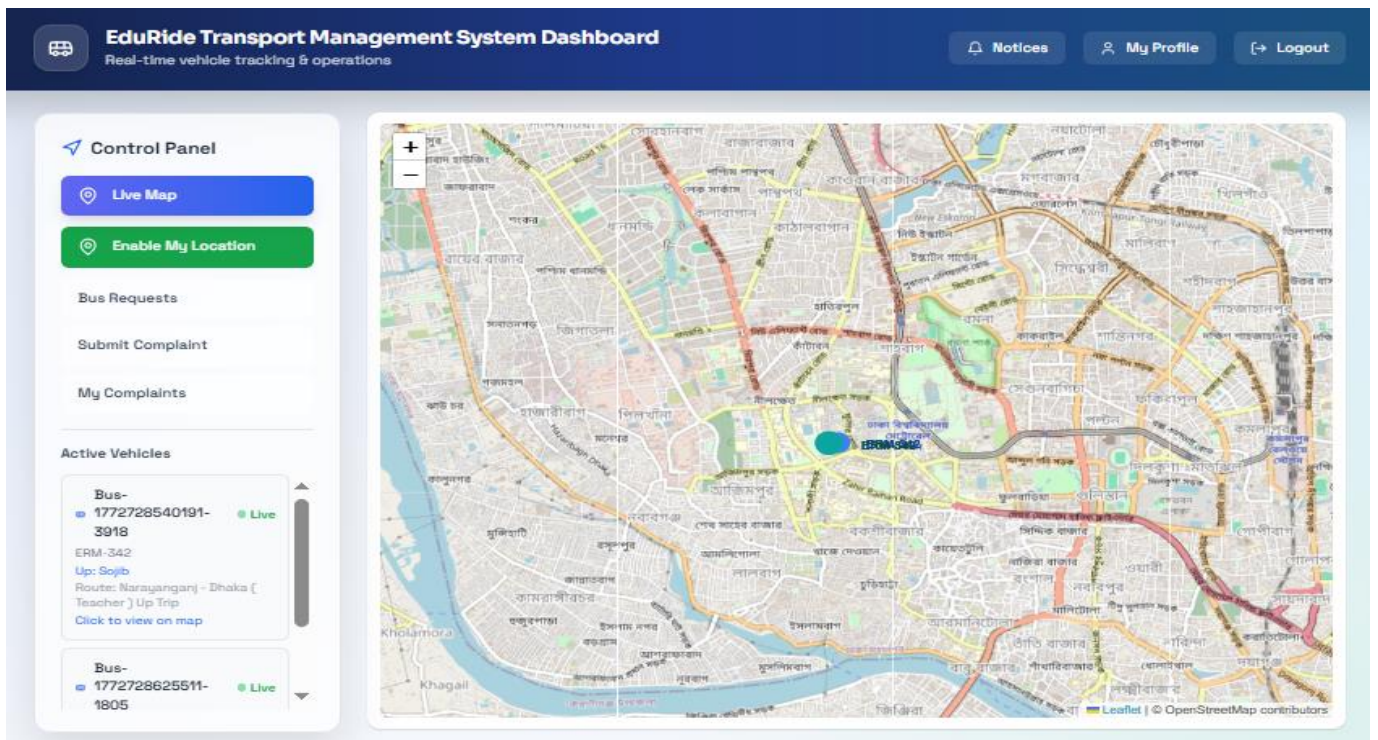


Figure 20: Teacher Dashboard (Live Map).

Satff Dashboard Interface

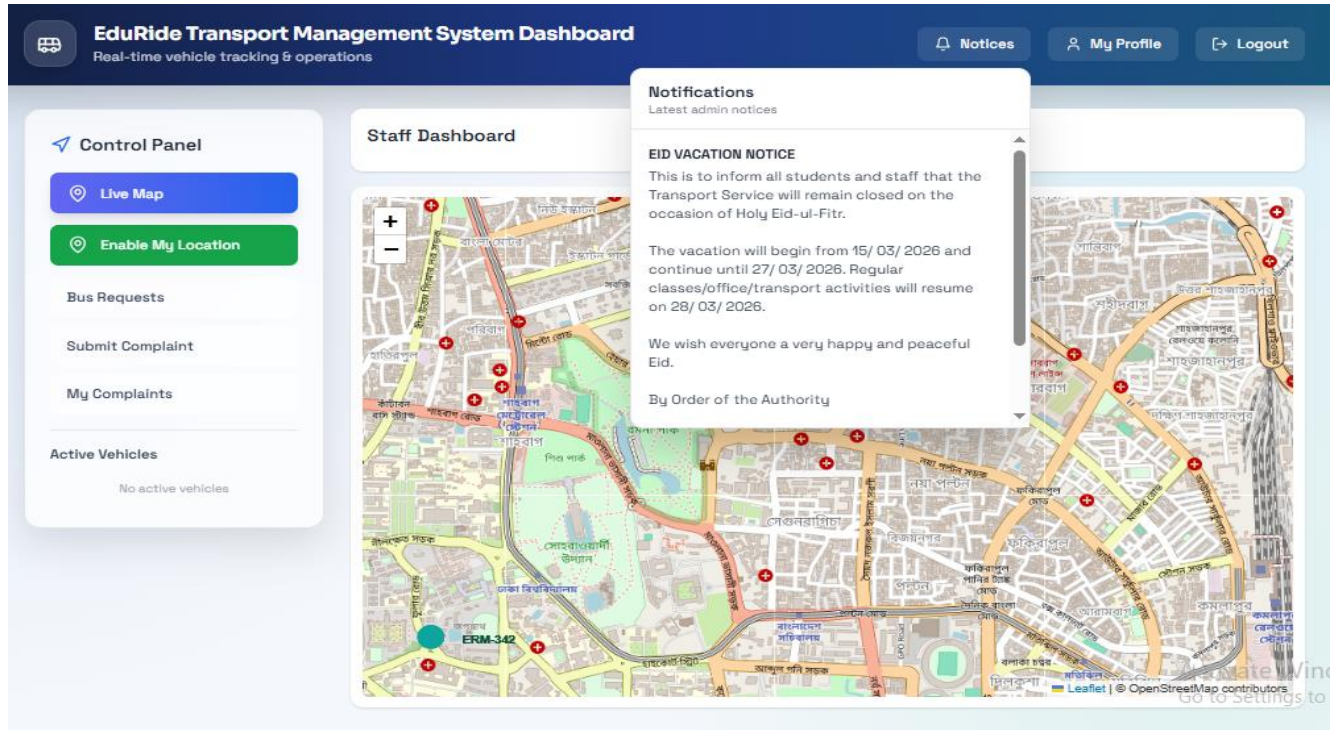


Figure 21: Staff Dashboard (Notice and Live Map).

Driver Dashboard Interface

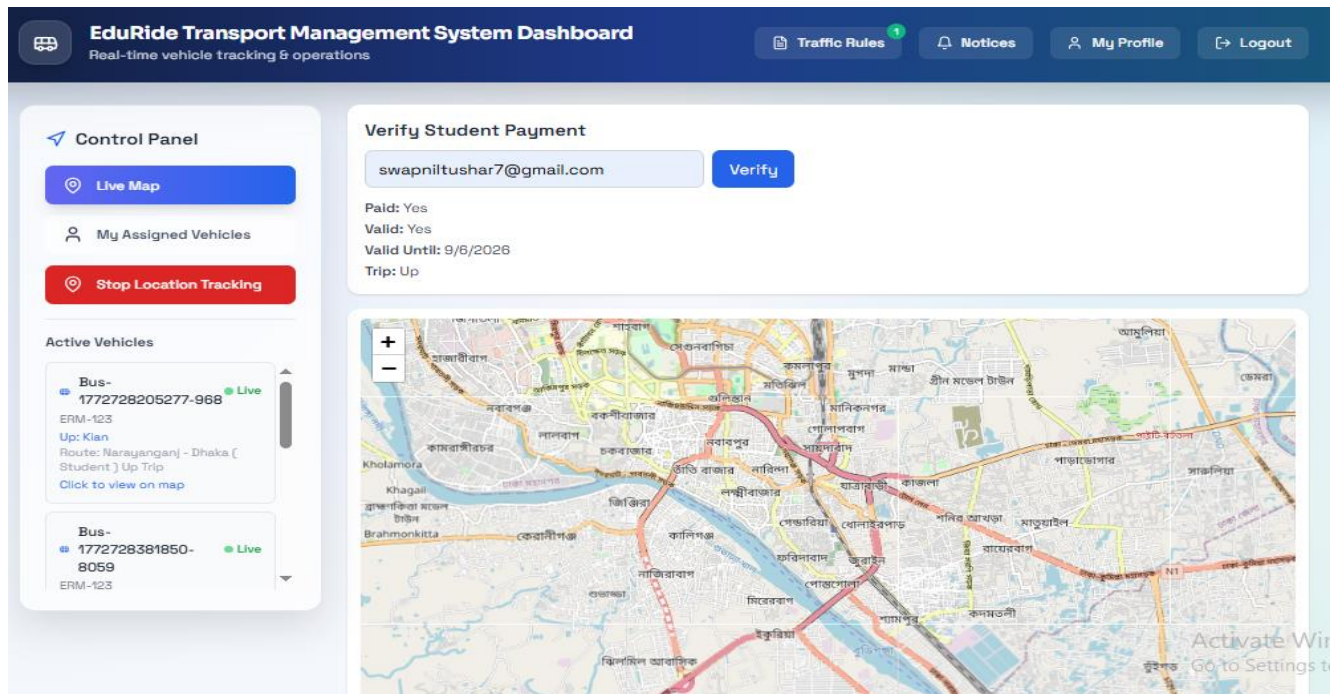


Figure 22: Driver Dashboard (Verify Student Payment and Live Map).

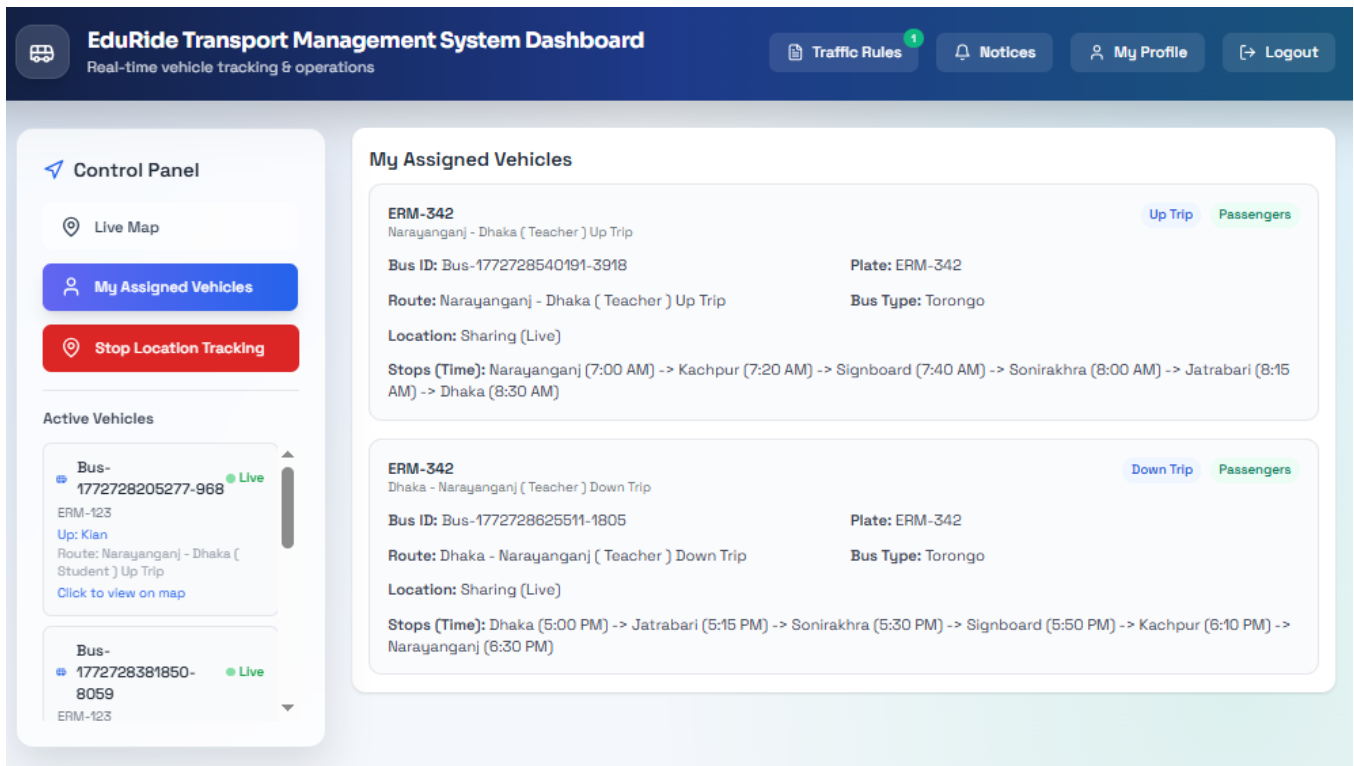


Figure 23: Driver Dashboard (My Assigned Vehicles).

5.2 MongoDB Collections

MongoDB Compass - cluster0.tbhw0sf.mongodb.net/test

Connections Edit View Help

Compass

My Queries
Data Modeling

CONNECTIONS (2)

Search connections

cluster0.tbhw0sf.mongodb.net

- admin
- config
- local
- sample_mflix
- test
 - buses
 - busrequests
 - complaints
 - notices
 - payments
 - routes
 - trafficrules
 - users
 - vehicles

cluster0.tbhw0sf.mongodb.net > test

Open MongoDB shell Create collection Refresh

Collection name	Properties	Storage size	Documents	Avg. document size	Indexes	Total index size
buses	-	397.31 kB	7	25.55 kB	2	73.73 kB
busrequests	-	36.86 kB	7	1.14 kB	1	36.86 kB
complaints	-	36.86 kB	1	269.00 B	2	73.73 kB
notices	-	36.86 kB	1	567.00 B	1	36.86 kB
payments	-	36.86 kB	7	407.00 B	2	73.73 kB
routes	-	36.86 kB	7	548.00 B	2	73.73 kB
trafficrules	-	36.86 kB	1	292.00 B	1	36.86 kB
users	-	393.22 kB	8	33.44 kB	2	73.73 kB
vehicles	-	77.82 kB	6	3.62 kB	2	73.73 kB

Activate Windows
Go to Settings to activate Windows.

Figure 24: Database Collection (EduRide Transport Management System).

Chapter 6

Testing and Results

6.1 Unit Testing

Test Case ID	Module Name	Test Description	Input	Expected Output	Status
UT-01	User Registration	Valid user registration check	Valid name, email, password	User created successfully	Pass
UT-02	Login Module	Correct credential login	Valid email & password	JWT token generated	Pass
UT-03	Login Module	Invalid password check	Wrong password	Error message displayed	Pass
UT-04	Fleet	Add new bus	Bus data valid	Bus saved in database	Pass
UT-05	Route Module	Create new route	Route details valid	Route stored successfully	Pass
UT-06	Complaint Module	Submit complaint	Complaint text	Complaint saved	Pass
UT-07	Payment Module	Payment success update	Valid payment ID	Payment status updated	Pass
UT-08	GPS Tracking	Enable location sharing	Click Enable location	Location updated in system	Pass
UT-09	Notice	Publish notice	Notice content	Notice visible to users	Pass
UT-10	Payment Receipt	Download receipt	Click button	Download receipt successfully	Pass
UT-11	Student Transport Card	Download card	Click button	Download transport card successfully	Pass
UT-12	Assignment	Assign driver to route	Driver Name + Route Name	Assignment stored	Pass

Table 1: Unit Testing of EduRide Transport Management System.

6.2 Integration Testing

Integration Scenario	Modules Involved	Test Objective	Expected Result	Status
User Login Flow	Authentication + Database	Verify user data retrieval	User logged in successfully	Pass
Route Selection with Approval	Student + Route Module + Admin	Check approval workflow	Admin can approve/reject/pending	Pass
Driver Assignment	Admin + Assignment + Database	Verify route-driver linkage	Driver assigned correctly	Pass
Live Tracking	Driver + GPS + Live Map	Check real-time tracking	Location visible on map	Pass
Online Payment Process	Student + Payment Gateway + Payment DB	Verify secure transaction	Payment stored & receipt and Card generated	Pass
Complaint Feedback Flow	Student/Teacher/Staff + Admin + Complaint Module	Check complaint submission & response	Feedback delivered successfully	Pass
Notice Publishing	Admin + Notice Module + All Users	Verify notice visibility	Notice visible to all roles	Pass
Vehicle Tracking	Fleet + GPS + Database	Ensure correct vehicle data sync	Correct vehicle location shown	Pass

Table 2: Integration Testing of EduRide Transport Management System.

6.3 Performance Testing

Module Name	Test Scenario	Load Condition	Expected Performance	Result
Authentication	Multiple user login	6 concurrent users	Response time < 3 sec	Stable
Live Tracking	Real-time GPS updates	3 buses updating	Location update within 5 sec	Stable
Payment System	Simultaneous payments	5 transactions	No transaction failure	Stable
Route Selection	Peak hour slot booking	6 users	No system crash	Stable
Complaint System	Complaint submission	1 submissions	Stored without delay	Stable
Notice Module	Mass notice access	6 users viewing	Page loads within 2 sec	Stable
System Overall	Full system load test	6 active users	No downtime	Stable

Table 3: Performance Testing of EduRide Transport Management System.

6.4 Testing Summary

The EduRide Transport Management System was tested through Unit, Integration, and Performance Testing.

Unit testing confirmed that all individual modules—such as registration, login, route management, payment, GPS tracking, complaints, and notice publishing—worked correctly.

Integration testing verified smooth communication between connected modules, including authentication with the database, route approval workflows, driver assignment, live tracking, and payment processing.

Performance testing showed that the system handled multiple users and transactions efficiently, maintaining response times under 2–3 seconds without crashes or downtime. Overall, the system is functional, stable, and ready for reliable campus transport management use.

Chapter 7

Conclusion and Future Work

7.1 Overview

The EduRide Transport Management System has been successfully developed to address the challenges associated with manual campus transport operations. This project aimed to automate transport tasks such as enrollment, route and slot allocation, real-time vehicle tracking, payment handling, complaints management, and communication through an integrated digital platform. Built using modern web technologies, the system demonstrates robust functionality, scalability, and improved user experience for all stakeholders including administrators, students, teachers, staff, and drivers.

Comprehensive testing—including unit, integration, and performance evaluations—was conducted to ensure that the system meets both functional and non-functional requirements. The system’s ability to handle multiple users, real-time GPS updates, and secure transactions has been verified under moderate load conditions, confirming its readiness for practical deployment in an educational environment.

7.2 Limitation of the Project

While EduRide significantly improves the transport management process, several limitations were identified during development and testing:

Dependency on Internet Connectivity:

Real-time GPS tracking and online payment features require stable internet access. In areas with poor network coverage, system responsiveness may degrade.

Driver Phone-Based Tracking:

Currently, location updates rely on the driver's mobile device. This approach may be affected by battery drain, device performance, or GPS accuracy variations.

Scalability Constraint in High Load Scenarios:

Although tested for moderate load, the system's performance under very high concurrent users (e.g., thousands of simultaneous tracking requests) needs further optimization.

User Role Dependencies:

Certain functions like slot approval and driver assignment depend on Admin intervention, which may introduce manual bottlenecks.

These limitations provide valuable insights for future improvement and system refinement.

7.3 Future Work

Several enhancements can be undertaken in future iterations to further strengthen EduRide's capabilities:

Nearby Vehicle Finder:

The Nearby Vehicle Finder will allow students, teachers, and staff to locate the closest available campus transport vehicle in real time using GPS technology and AI.

Mobile Application Development:

Develop companion mobile applications for Android and iOS to improve the accessibility and usability of the platform for students and drivers alike.

IoT-Based Vehicle Health Monitoring:

Connect sensors to monitor Fuel level, Engine condition, Tire pressure. Send maintenance alerts notification and information to Admin dashboard before breakdown occurs.

Advanced Analytics and Reporting:

Add analytical dashboards featuring insights such as route usage, peak demand periods, and driver performance metrics to help administrators make data-driven decisions.

7.4 Conclusion

The EduRide Transport Management System delivers an effective and scalable solution for managing campus transportation services digitally. By replacing manual and paper-based systems with a centralized platform, the project enhances transparency, reduces administrative workloads, and improves overall service quality for all users. The system successfully integrates core functionalities such as user management, real-time tracking, payment processing, complaint handling, and notice distribution into a unified application. Through rigorous testing, EduRide proved to be both functionally reliable and performance-stable under expected operational conditions. Despite certain limitations, the project sets a strong foundation for future development and real-world implementation. With proposed enhancements such as dedicated GPS hardware, mobile apps, and advanced analytics, EduRide has the potential to evolve into a comprehensive transport solution suitable for larger institutional needs. Overall, this project demonstrates a significant contribution toward modernizing campus transportation management with modern web technologies.

References

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